

Sai Yadav Banda

Python Full Stack Developer

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PROFESSIONAL SUMMARY:

- Python Full Stack Developer with 4+ years of extensive experience in designing, developing and deploying high-performance, scalable and secure applications within demanding FinTech and e-commerce environments.
- Expert in architecting resilient microservices using Django, FastAPI and Flask and building real-time trading APIs (REST, GraphQL, gRPC) capable of sub-millisecond latency for critical financial operations.
- Adept at engineering sophisticated data pipelines with Apache Airflow and Snowflake, optimizing PostgreSQL and TimescaleDB schemas for large-scale time-series data and developing advanced fraud detection and risk calculation models leveraging Scikit-learn and XGBoost.
- Proficient in crafting dynamic, low-latency front-end experiences using React 18, Next.js 14, Redux Toolkit and React Query, ensuring intuitive user interfaces with Material UI and Tailwind CSS.
- Skilled in implementing robust security measures including PCI DSS and SOC2 compliance, end-to-end encryption (KMS/HSM), JWT authentication and API Gateway management.
- Experienced in automating CI/CD pipelines with GitHub Actions and Jenkins, managing containerized deployments with AWS Elastic Beanstalk and Kubernetes (EKS) and ensuring comprehensive test coverage with PyTest and Cypress for mission-critical systems.

TECHNICAL SKILLS:

Programming Languages:	Python (3.12+), JavaScript (ES2023+), TypeScript 5.x, SQL, HTML5, CSS3, SCSS
Frameworks & Libraries:	Django, Django REST Framework (DRF), FastAPI, Flask, react 18+, Next.js 14, Redux Toolkit, React Query, Material UI, Tailwind CSS, Cypress
API Design & Protocols:	RESTful APIs, GraphQL, gRPC, WebSockets, FIX Protocol, SWIFT Messaging, ISO 20022, OAuth2, JWT Authentication
Databases & Data Stores:	PostgreSQL, TimescaleDB, MongoDB, Snowflake, Redis, Elasticsearch
Data Engineering & ETL:	Apache Airflow, Kafka, ETL Pipelines, Pandas, NumPy, XGBoost, Scikit-learn
Cloud Platforms & DevOps:	AWS (EC2, S3, RDS, Elastic Beanstalk, Cognito, API Gateway, KMS, HSM), Kubernetes (EKS), Docker, Jenkins, GitHub Actions, CI/CD Pipelines, Terraform
Big Data & Streaming:	Apache Kafka, WebSocket Streaming, Time-Series Data Processing
Security & Compliance:	PCI DSS, SOC2, End-to-End Encryption, AES Encryption, Secure Key Management (KMS/HSM), API Gateway Rate Limiting, Access Control (RBAC), OAuth2, HTTPS/TLS
Testing & QA:	PyTest, Unit Testing, Integration Testing, Cypress, Test-Driven Development (TDD), Continuous Testing
Monitoring & Logging:	Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), AWS CloudWatch
Version Control & Tools:	Git, GitHub, Bitbucket, JIRA, Confluence, Agile/Scrum Methodologies

PROFESSIONAL EXPERIENCE:

KPMG US – GA
Python Developer

April 2024 – Present

Roles and Responsibilities:

- Implemented scalable microservices using Django, FastAPI and Flask aligned with Domain-Driven Design (DDD) principles, ensuring modularity and fault isolation across trading applications.
- Built high-performance real-time trading APIs (REST, GraphQL, gRPC) capable of processing 50K+ transactions/sec with sub-millisecond latency, directly improving trade execution speed by 45%.
- Designed WebSocket-based streaming pipelines for market data feeds and order book synchronization, enabling real-time trading decisions and reducing latency in price discovery.
- Integrated FIX protocol, SWIFT messaging and ISO 20022 standards for secure interbank settlements, ensuring compliance with international banking regulations.

- Developed fraud detection and anomaly detection ML models using Scikit-learn and XGBoost integrated with Kafka-driven event streams, reducing false positives by 18% and fraudulent transaction losses by 28%.
- Created risk calculation engines for derivatives trading using Python, NumPy and Pandas, ensuring accurate Value-at-Risk (VaR) computations and adherence to regulatory reporting standards.
- Engineered optimized ETL pipelines using Apache Airflow and Snowflake to aggregate and analyze historical trading data, improving data pipeline processing efficiency by 65%.
- Designed PostgreSQL and TimescaleDB schemas optimized for time-series financial data, cutting complex query times by 40% and improving analytics capabilities for large-scale datasets.
- Developed React 18 + Next.js 14 frontends for trading dashboards with Redux Toolkit and React Query for real-time state management, delivering low-latency visualization of market and risk metrics.
- Ensured PCI DSS and SOC2 compliance through end-to-end encryption (KMS/HSM), JWT-based authentication and API Gateway-based rate limiting, enhancing platform security and maintaining 99.98% uptime.

Capgemini – India
Python Developer

June 2021 – July 2023

Roles and Responsibilities:

- Designed and developed RESTful APIs using Django REST Framework to manage product catalogs, user authentication and order processing, ensuring seamless integration with e-commerce workflows.
- Built responsive front-end interfaces with React.js, Material UI and Tailwind CSS, delivering intuitive and device-agnostic user experiences for online retail platforms.
- Increased online sales by 40% by optimizing user experience and reducing page load times through Next.js and Redis caching strategies.
- Reduced order processing time by 30% by implementing Apache Kafka for real-time synchronization of orders and inventory across the platform.
- Improved customer retention by 25% through personalized product recommendation systems built with Pandas and Elasticsearch for efficient data handling and search.
- Integrated payment gateways like Razorpay and Stripe, ensuring PCI DSS compliance and securing sensitive data with AES encryption.
- Developed real-time shipping tracking by integrating Delhivery and Shippo APIs, enhancing delivery transparency and customer satisfaction.
- Configured and optimized databases using PostgreSQL and MongoDB for efficient storage and retrieval of product, user and transaction data.
- Enhanced platform scalability to handle high concurrent user traffic during peak sales using AWS Elastic Beanstalk and Kubernetes (EKS).
- Automated CI/CD pipelines with GitHub Actions and Jenkins, enabling faster and reliable deployment cycles for e-commerce applications.
- Wrote comprehensive unit and integration tests using PyTest and Cypress, achieving 90% code coverage to ensure robust application performance.
- Implemented secure authentication with OAuth2 and JWT via AWS Cognito, ensuring safe and scalable user access management for retail platforms.

EDUCATION:

Masters in Technology Management - Lindsey Wilson College, Bowling Green, Kentucky, USA

Bachelor of Commerce - Acharya Nagarjuna University, Guntur, Andhra Pradesh, India